SESSION 14 – K-NEAREST NEIGHBORS (KNN – CLASSIFICATION)

🎯 Topic

- What is KNN
- Distance-based learning
- Role of "K"
- How prediction works
- Importance of scaling
- Practical implementation

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1. 🎯 Learning Objectives

By the end of this session, students will be able to:

- Understand how **KNN works**
- Learn concept of **distance-based classification**
- Understand role of **K (number of neighbors)**
- Know importance of **feature scaling**
- Build a **KNN classification model**
- Interpret predictions

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2. 🔁 Quick Recap (Connect Previous Session)

Start class like this:

"Till now we used models like Logistic Regression and Decision Trees.
Today we will learn a very simple idea — instead of building a formula, we look at nearby data points to make decisions."

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3. 📖 What is KNN?

3.1 Definition

K-Nearest Neighbors (KNN) is a supervised learning algorithm that classifies data based on the **nearest data points**.

3.2 Simple Meaning (Most Important Line)

"KNN predicts the class by looking at the nearest neighbors."

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4. 📊 How KNN Works

Step-by-Step

1. Choose value of K
2. Calculate distance from new point to all data points
3. Select K nearest neighbors
4. Take majority vote
5. Assign class

Teaching Line

"KNN learns from neighbors, not from equations."

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5. 📊 Example (Very Important)

Pass/Fail Example

New student:

- Study Hours = 5

Check nearest neighbors:

- 3 Pass
- 2 Fail

👉 Prediction:

Pass (majority)

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6. 📏 Distance Calculation

Common Distance: Euclidean

$$Distance = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Teaching Line

"Closer points have more influence on prediction."

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7. 📊 Role of K (Very Important)

Small K (e.g., 1)

- Very sensitive

- Can overfit

Large K (e.g., 10)

- Smooth prediction
- Can underfit

Teaching Line

“Choosing the right K is very important.”

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8. 📊 Odd vs Even K

👉 Prefer **odd K**

Why?

- Avoid tie

Example:

- K = 3 → clear majority
- K = 4 → possible tie

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9. ⚠️ Importance of Feature Scaling

Problem

If features have different scales:

Feature	Range
Study Hours	1–10
Salary	10,000–100,000

👉 Distance becomes biased

Solution

👉 Apply scaling:

- StandardScaler
- MinMaxScaler

Teaching Line

“KNN is very sensitive to feature scale.”

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10. 📊 Advantages

✅ Simple to understand
✅ No training time
✅ Works well for small datasets

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11. ⚠️ Limitations

❌ Slow for large data
❌ Sensitive to noise
❌ Requires scaling

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12. 🎯 Interview Concepts

Q1: What is KNN?

👉 Distance-based classification algorithm.

Q2: What is role of K?

👉 Number of neighbors used for prediction.

Q3: Why scaling is important?

👉 To avoid bias in distance calculation.

Q4: What happens if K is too small?

👉 Overfitting.

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13. 💻 PRACTICAL (YOUR FORMAT)

🎯 Objective

Predict Pass/Fail using KNN

14. 📊 Output Interpretation

Explain:

- Model checks nearest neighbors

- Uses majority voting
- Scaling ensures fair distance

👉 Example:

- Closest students → mostly Pass
  👉 Output → Pass

KNN is one of the simplest yet powerful algorithms.
It makes decisions based on similarity and distance, just like humans compare with similar cases.